

AMC8 数学竞赛 第 1-5 题

1. What is the ones digit of? 求下列算式结果的个位上的数字。(计算：基本运算)

$$222,222 - 22,222 - 2,222 - 222 - 22 - 2?$$

- (A) 0 (B) 2 (C) 4 (D) 6 (E) 8

【答案】B

【解析】

$$\begin{array}{r} 22222 \\ + 2222 \\ \quad 222 \\ \quad \quad 22 \\ \quad \quad \quad 2 \\ \hline \dots 0 \end{array} \quad \begin{array}{r} 222222 \\ - \dots 0 \\ \hline \dots 2 \end{array} \quad \text{尾数是 2}$$

2. What is the value of this expression in decimal form? 把下列算式的结果写成小数形式是单位?

$$\frac{44}{11} + \frac{110}{44} + \frac{44}{1100}$$

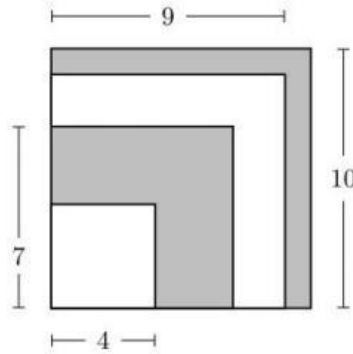
- (A) 6.4 (B) 6.504 (C) 6.54 (D) 6.9 (E) 6.94

【答案】C

【解析】

$$\begin{aligned} & 4 + \frac{5}{2} + \frac{4}{100} \\ &= 4 + 2.5 + 0.04 \\ &= 6.54 \end{aligned}$$

3. Four squares of side length 4, 7, 9, and 10 units are arranged in increasing size order so that their left edges and bottom edges align. The squares alternate in color white-gray-white-gray, respectively, as shown in the figure. What is the area of the visible gray region in square units? 四个边长分别为 4、7、9 和 10 单位的正方形按边长递增的顺序排列，使得它们的左边缘和底边缘对齐。正方形交替填充为白色和灰色，如下图所示。灰色部分的面积是多少平方



- (A) 42 (B) 45 (C) 49 (D) 50 (E) 52

【答案】E

【解析】

$$\begin{aligned}
 &7^2 - 4^2 + 10^2 - 9^2 \\
 &= (7+4)(7-4) + (10-9)(10+9) \\
 &= 11 \times 3 + 19 = 52
 \end{aligned}$$

3. When Yunji added all the integers from 1 through 9, she mistakenly left out a number. Her incorrect sum turned out to be a square number. Which number did Yunji leave out? 当 Yunji 从 1 加到 9 的时候, 不小心遗漏了一个数字。使得错误的和是一个平方数。那么 Yunji 遗漏的数字是多少?

- (A) 5 (B) 6 (C) 7 (D) 8 (E) 9

【答案】E

【解析】

$$1+2+3+\cdots+9=45$$

少一个, 范围在 36 ~ 45 之间

只有 36 是平方数, 少加了 $45-36=9$

5. Aaliyah rolls two standard 6-sided dice. She notices that the product of the two numbers rolled is a multiple of 6. Which of the following integers cannot be the sum of the two numbers?

Aaliyah 掷了两个标准六面骰子。她发现掷出的两个数字的乘积是 6 的倍数。以下哪个整数不能是这两个数字的和?

- (A) 5 (B) 6 (C) 7 (D) 8 (E) 9

【答案】B

【解析】

$$A: 5: 2+3 \checkmark$$

$$C: 7: 3+4 \checkmark$$

$$D: 8: 2+6 \checkmark$$

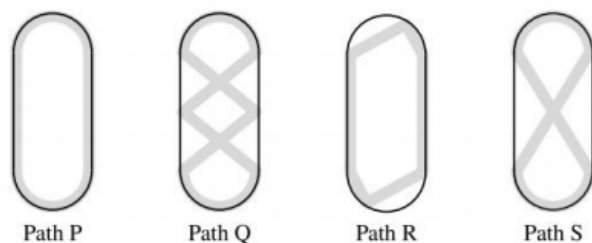
$$E: 9: 3+6 \checkmark$$

$$B: 6: 1+5 \times$$

$$2+4 \times$$

$$3+3 \times$$

6. Sergei skated around an ice rink, gliding along different paths. The gray lines in the figures below show four of the paths labeled P, Q, R, and S. What is the sorted order of the four paths from shortest to longest? 谢尔盖在溜冰场周围滑冰，沿着不同的路径滑行。以下图形中，灰色线条显示了四条路径 P、Q、R 和 S。按照从最短到最长的顺序，这四条路径要如何排序？



- (A) P, Q, R, S (B) P, R, S, Q (C) Q, S, P, R (D) R, P, S, Q (E) R, S, P, Q

【答案】D

【解析】

根据弯曲程度比较：R 最直
P 其次
S 有部分弯曲
Q 最多弯曲

7. A 3×7 rectangle is covered without overlap by 3 shapes of tiles: 2×2 , 1×4 , and 1×1 , shown below. What is the minimum possible number of 1×1 tiles used?

一个 3×7 的矩形被无重叠地覆盖了 3 种形状的瓷砖： 2×2 、 1×4 和 1×1 。如下图所示。使用的 1×1 瓷砖的最少可能数量是多少？



- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

【答案】E

【解析】

0	0	0	0	0	0	空
0	0	0	0	0	0	空
0	0	0	0	空	空	空

共5个

8. On Monday Tye has \$2. Every day, he either gains \$3 or doubles the amount of money he had on the previous day. How many different dollar amounts could Tye have on Thursday 3 days later? 在星期一, Tye 有 2 美元。每天, 他要么获得 3 美元, 要么将前一天的钱数翻倍。那么在三天后的星期四, Tye 可能拥有多少不同的美元总金额?

(A) 3 (B) 4 (C) 5 (D) 6 (E) 7

【答案】D

【解析】

$$\begin{array}{ll}
 2+3+3+3=11 \checkmark & (2+3) \times 2+3=13 \checkmark \\
 2 \times 2 \times 2 \times 2=16 \checkmark & 2 \times 2+3+3=10 \checkmark \\
 (2+3) \times 2 \times 2=20 \checkmark & \text{分类 + 分支} \\
 (2 \times 2+3) \times 2=14 \checkmark & \\
 2 \times 2 \times 2+3=11 \times & \\
 (2+3+3) \times 2=16 \times &
 \end{array}$$

9. All of the marbles in Maria's collection are red, green, or blue. Maria has half as many red marbles as green marbles and twice as many blue marbles as green marbles. Which of the following could be the total number of marbles in Maria's collection? 玛丽亚收集的弹珠都是红色、绿色或蓝色的。玛丽亚拥有的红色弹珠数量是绿色弹珠的一半, 拥有的蓝色弹珠数量是绿色弹珠的两倍。下面哪个可能是玛丽亚收集的弹珠总数?

(A) 24 (B) 25 (C) 26 (D) 27 (E) 28

【答案】E

【解析】

$$\begin{array}{l}
 R, G, B \\
 R = \frac{1}{2}G \Rightarrow G = 2R \\
 B = 2G = 4R \\
 \text{总数为 } 7 \text{ 的倍数} \\
 \therefore \text{一共 } 7R \Rightarrow 28
 \end{array}$$

10. In January 1980 the Mauna Loa Observatory recorded carbon dioxide (CO₂) levels of 338ppm (parts per million). Over the years the average CO₂ reading has increased by about 1.515 ppm each year. What is the expected CO₂ level in ppm in January 2030? Round your answer to the nearest integer. 1980 年 1 月，莫纳罗亚天文台记录的二氧化碳 (CO₂) 水平为 338ppm (百万分之一)。多年来，平均 CO₂ 读数每年增加约 1.515 ppm。2030 年 1 月的预期 CO₂ 水平是多少 ppm? 四舍五入到最接近的整数。

- (A) 399 (B) 414 (C) 420 (D) 444 (E) 459

【答案】B

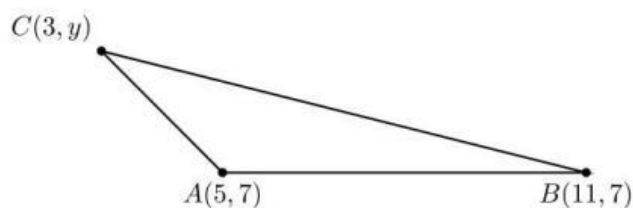
【解析】

$$50 \times 1.515 = 75.75$$

$$338 + 75.75 \approx 414$$

AMC8 数学竞赛 第 11-15 题

11. The coordinates of $\triangle ABC$ are $A(5, 7)$, $B(11, 7)$ and $C(3, y)$, with $y > 7$. The area of $\triangle ABC$ is 12. What is the value of y ? 三角形 ABC 的坐标分别为 $A(5, 7)$ 、 $B(11, 7)$ 和 $C(3, y)$ ，其中 $y > 7$ 。三角形 ABC 的面积为 12，求 y 的值。



- (A) 8 (B) 9 (C) 10 (D) 11 (E) 12

【答案】D

【解析】

$$\text{底 } AB = 11 - 5 = 6$$

$$S = \frac{1}{2} \times 6 \times h = 12$$

$$h = 4 = y - 7$$

$$\therefore y = 11$$

12. Rohan keeps a total of 90 guppies in 4 fish tanks.

- There is 1 more guppy in the 2nd tank than in the 1st tank.
- There are 2 more guppies in the 3rd tank than in the 2nd tank.
- There are 3 more guppies in the 4th tank than in the 3rd tank.

How many guppies are in the 4th tank?

罗罕总共在 4 个鱼缸里养了 90 条孔雀鱼。

- 第二个鱼缸里的孔雀鱼比第一个鱼缸里的多 1 条。
- 第三个鱼缸里的孔雀鱼比第二个鱼缸里的多 2 条。
- 第四个鱼缸里的孔雀鱼比第三个鱼缸里的多 3 条。

第四个鱼缸里有多少条孔雀鱼？

- (A) 20 (B) 21 (C) 23 (D) 24 (E) 26

【答案】E

【解析】

设第一个缸有 x 条, 接下来依次有: $x+1$ $x+3$ $x+6$

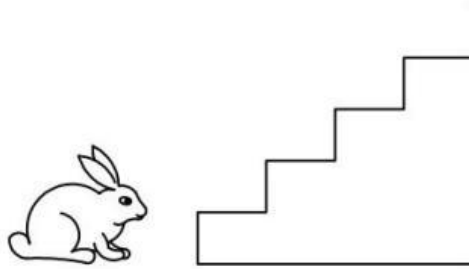
$$90 = x + x + 1 + x + 3 + x + 6$$

$$90 = 4x + 10$$

$$x = 20$$

$$\therefore x + 6 = 26$$

13. Buzz Bunny is hopping up and down a set of stairs, one step at a time. In how many ways can Buzz start on the ground, make a sequence of 6 hops, and end up back on the ground? (For example, one sequence of hops is up-up-down-down-up-down.) Buzz Bunny 一次只能跳一个台阶, 正在上下跳跃。问 Buzz 有几种方式可以从地面开始, 跳跃 6 次, 最终回到地面? (例如, 其中一种跳跃序列是上、上、下、下、上、下。)



(A) 4 (B) 5 (C) 6 (D) 8 (E) 12

【答案】B

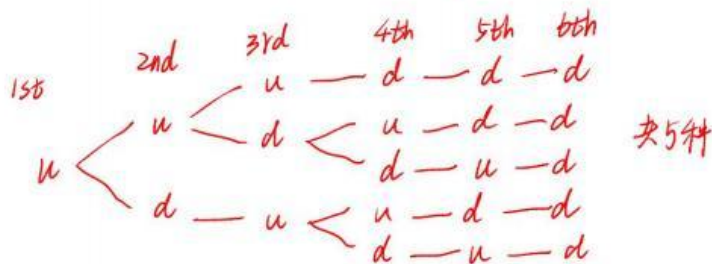
【解析】

根据条件, 兔子总共跳 6 次, 最后回到原地,

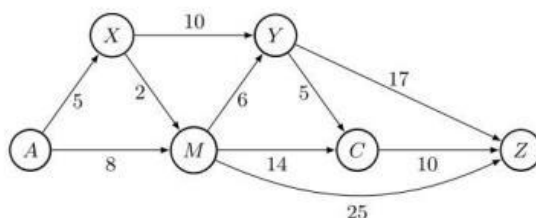
$\Rightarrow$ 往上和往下的次数相等, 各为 3 次

题目隐含限制条件较多, 并且跳跃的总次数较少

选择用树状图列举: ($u = up$, $d = down$)



14. The one-way routes connecting towns A, M, C, X, Y, and Z are shown in the figure below (not drawn to scale). The distances in kilometers along each route are marked. Traveling along these routes, what is the shortest distance from A to Z in kilometers? 连接城镇 A、M、C、X、Y、Z 的单向路线如下图所示（未按比例绘制）。沿每条路线的距离以公里为单位进行标记。沿着这些路线旅行所有城镇都要去一遍，A 到 Z 之间最短的距离是多少公里？



- (A) 28 (B) 29 (C) 30 (D) 31 (E) 32

【答案】A

【解析】

从 A 到 Z 的最短路线为 $A \rightarrow X \rightarrow M \rightarrow Y \rightarrow C \rightarrow Z$
 总路程为 $5 + 2 + 6 + 5 + 10 = 28$

15. Let the letters F, L, Y, B, U, G represent distinct digits. Suppose FLYFLY is the greatest number that satisfies the equation:

$$8 \cdot \text{FLYFLY} = \text{BUGBUG}$$

What is the value of FLY + BUG?

用 F, L, Y, B, U, G 表示不同的数字。假设 FLYFLY 是满足方程 $8 \times \text{FLYFLY} = \text{BUGBUG}$ 的最大数字，求 FLY + BUG 的值。

- (A) 1089 (B) 1098 (C) 1107 (D) 1116 (E) 1125

【答案】C

【解析】

$$\begin{aligned} 8 \cdot \text{FLYFLY} &= \text{BUGBUG} \\ 8 \cdot 1001 \cdot \text{FLY} &= 1001 \cdot \text{BUG} \\ 8 \cdot \text{FLY} &= \text{BUG} \end{aligned}$$

$$8 \text{ 与一个三位数相乘仍为三位数} \Rightarrow \text{FLY} < \frac{1000}{8} = 125$$

$$\text{当 } \text{FLY} = 124 \text{ 时, } \text{BUG} = 8 \times 124 = 992, \text{ 不满足条件}$$

$$\text{当 } \text{FLY} = 123 \text{ 时, } \text{BUG} = 8 \times 123 = 984, \text{ 满足条件}$$

$$\text{FLY} + \text{BUG} = 123 + 984 = 1107$$

AMC8 数学竞赛 第 16-20 题

16. Minh enters the numbers 1 through 81 into the cells of a 9×9 grid in some order. She calculates the product of the numbers in each row and column. What is the least number of rows and columns that could have a product divisible by 3?

Minh 将数字 1 到 81 按照某种顺序放入 9×9 网格的单元格中。她计算每一行和每一列中数字的乘积。乘积可被 3 整除的行数和列数最少为多少？

- (A) 8 (B) 9 (C) 10 (D) 11 (E) 12

【答案】D

【解析】

根据数论知识,若某行或某列所有数的乘积能被 3 整除,则该行

(列)中的数至少有 1 个 3 的倍数

1-81 中 3 的倍数共有 $\lfloor \frac{81}{3} \rfloor = 27$ 个

要让乘积能被 3 整除的行或列最少,则应让 3 的倍数分布的行和列最少,最少情况如图:

x	x	x	x	x	x		
x	x	x	x	x	x		
x	x	x	x	x	x		
x	x	x	x	x			

4 行 7 列

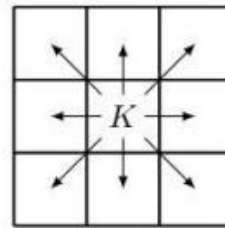
$$4+7=11$$

17. A chess king is said to attack all the squares one step away from it, horizontally, vertically, or diagonally. For instance, a king on the center square of a 3×3 grid attacks all 8 other squares, as shown below. Suppose a white king and a black king are placed on different squares of a 3×3 grid so that they do not attack each other. In how many ways can this be done?

据说国际象棋的国王可以攻击其横向、纵向或对角线上相隔一步的所有空格。

例如，位于 3×3 网格中心的国王可以攻击其他 8 个空格，如下所示。

假设白王和黑王被放置在 3×3 网格的不同方格中，并且他们按照箭头方向移动一格也无法碰到对方，它们就不会互相攻击。有多少种方法可以做到这一点？



- (A) 20 (B) 24 (C) 27 (D) 28 (E) 32

【答案】E

【解析】

1	2	3
4	5	6
7	8	9

case 1: 若 Black king 在位置 1/3/7/9 任一位置, white king 都有 5 个位置

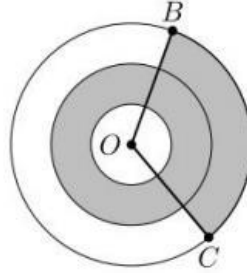
$$4 \times 5 = 20$$

case 2: 若 Black king 在 2/4/6/8 任一位置, white king 都有 3 个位置

$$4 \times 3 = 12$$

$$20 + 12 = 32$$

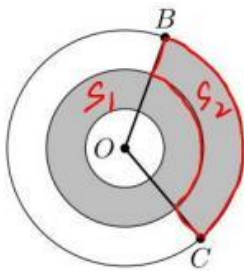
18. Three concentric circles centered at O have radius of 1, 2, and 3. Points B and C lie on the largest circle. The region between the two smaller circles is shaded, as is the portion of the region between the two larger circles bounded by central angle BOC , as shown in the figure below. Suppose the shaded and unshaded regions are equal in area, What is the measure of $\angle BOC$ in degrees? 以 O 为中心的三个同心圆的半径分别为 1、2 和 3。点 B 和点 C 在最大的圆上。两个较小的圆之间的区域被着色，两个较大的圆之间 BOC 角的所夹的被着色的部分，如下图所示。假设着色区域和未着色区域面积相等，则 $\angle BOC$ 的度数是多少？



- (A) 108 (B) 120 (C) 135 (D) 144 (E) 150

【答案】A

【解析】



根据条件, $S_{\text{阴}} = \frac{\pi \cdot 3^2}{2} = \frac{9\pi}{2}$

$$S_{\text{阴}} = S_1 + S_2$$

$$S_1 = \pi \cdot 2^2 - \pi \cdot 1^2 = 3\pi$$

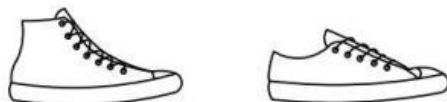
$$S_2 = S_{\text{阴}} - S_1 = \frac{9\pi}{2} - 3\pi = \frac{3\pi}{2}$$

S_2 占最外层圆环比值为:

$$\frac{\frac{3\pi}{2}}{\pi \cdot 3^2 - \pi \cdot 2^2} = \frac{\frac{3\pi}{2}}{5\pi} = \frac{3}{10}$$

$$\text{则 } \angle BOC = \frac{3}{10} \times 360^\circ = 108^\circ$$

19. Jordan owns 15 pairs of sneakers. Three fifths of the pairs are red and the rest are white. Two thirds of the pairs are high-top and the rest are low-top. The red high-top sneakers make up a fraction of the collection. What is the least possible value of this fraction? 乔丹有 15 双运动鞋。五分之三是红色的，其余都是白色的。三分之二是高帮的，其余都是低帮的。红色高帮运动鞋占总数的一小部分。这个分数的最小可能值是多少？



- (A) 0 (B) $\frac{1}{5}$ (C) $\frac{4}{15}$ (D) $\frac{1}{3}$ (E) $\frac{2}{5}$

【答案】c

【解析】

根据条件, $red = 15 \times \frac{3}{5} = 9$ $white: 15 - 9 = 6$

$high-top: 15 \times \frac{2}{3} = 10$ $low-top: 15 - 10 = 5$

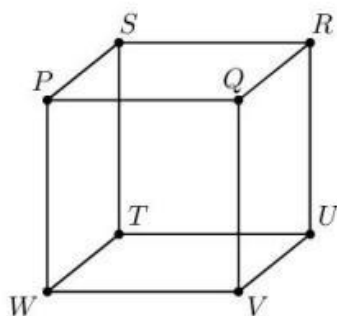
若要使 red high top 最少, 则 white 全为 high-top.

此时 red high top 为 $10 - 6 = 4$, 占比为 $\frac{4}{15}$

20. Any three vertices of the cube PQRSTUVW, shown in the figure below, can be connected to form a triangle. (For example, vertices P, Q, and R can be connected to form isosceles $\triangle POR$.) How many of these triangles are equilateral and contain P as a vertex?

如下图所示, 立方体 PQRSTUVW 的任意三个顶点都可以连接成一个三角形。(例如, 顶点 P、Q 和 R 可以连接成等腰三角形 $\triangle POR$ 。)这些三角形中, 有多少个是等边三角形并且

包含 P 作为一个顶点？



- (A) 0 (B) 1 (C) 2 (D) 3 (E) 6

【答案】D

【解析】

易知：边长为1和 $\sqrt{2}$ 的等边 Δ 不存在

$\Rightarrow$ 寻找连接为 $\sqrt{2}$ 的点

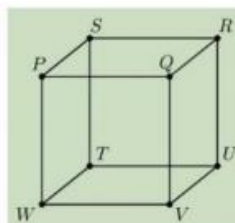
如右图，为 P、R、T、V (在两组对面)

$\Rightarrow$ 从这4点，任意选3点，可为等边 Δ

$$\Rightarrow C_4^3 = C_4^1 = 4$$

注意到：ARTV 不含 P

$$\Rightarrow N = 4 - 1 = 3$$



21. A group of frogs (called an army) is living in a tree. A frog turns green when in the shade and turns yellow when in the sun. Initially the ratio of green to yellow frogs was 3 : 1. Then 3 green frogs moved to the sunny side and 5 yellow frogs moved to the shady side. Now the ratio is 4 : 1. What is the difference between the number of green frogs and yellow frogs now? 一组青蛙（称为军队）住在树上。在阴凉处的青蛙会变绿，在阳光下则会变成黄色。最初，绿蛙与黄蛙的比例是 3 : 1。然后三只绿蛙移到阳光明媚的一侧，五只黄蛙移到阴凉处。现在的 ratio 是 4 : 1。现在绿蛙和黄蛙的数量差是多少？

- (A) 10 (B) 12 (C) 16 (D) 20 (E) 24

【答案】E

【解析】

设 $G = 3x$ ，则 $Y = x$

$$\Rightarrow \frac{3x - 3 + 5}{x - 5 + 3} = \frac{4}{1}$$

$$\Rightarrow 4x - 8 = 3x + 2$$

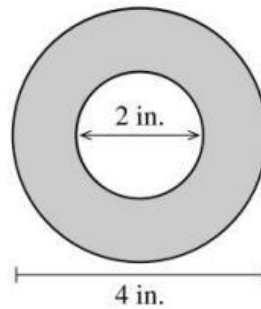
$$x = 10$$

$$3x + 2 = 32, \quad x - 2 = 8$$

$$\Rightarrow 32 - 8 = 24$$

22. A roll of tape is 4 inches in diameter and is wrapped around a ring that is 2 inches in diameter. A cross section of the tape is shown in the figure below. The tape is 0.015 inches thick. the tape is completely unrolled, approximately how long would it be? Round your answer the nearest 100 inches.

一卷磁带的直径为 4 英寸，并缠绕在直径为 2 英寸的环上。如下图所示是磁带的横截面。磁带的厚度为 0.015 英寸。如果完全展开磁带，它大约有多长？将答案四舍五入到最接近的 100 英寸。



- (A) 300 (B) 600 (C) 1200 (D) 1500 (E) 1800

【答案】B

【解析】

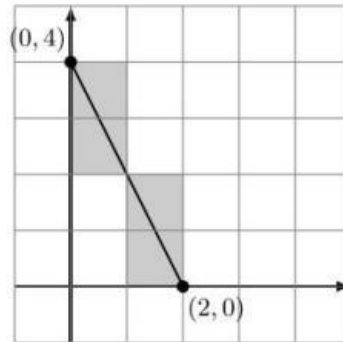
$$\frac{(2\pi + 4\pi)}{2} = 3\pi$$

$$3\pi \times \frac{1}{0.015} = \frac{1000}{15} \times 3\pi = \frac{1000}{5} \pi = 200\pi$$

$$\approx 600$$

23. Rodrigo has a very large piece of graph paper. First he draws a line segment connecting point $(0, 4)$ to point $(2, 0)$ and colors the 4 cells whose interiors intersect the segment, as shown below. Next Rodrigo draws a line segment connecting point $(2000, 3000)$ to point $(5000, 8000)$. again he colors the cells whose interiors intersect the segment. How many cells will he color this time?

Rodrigo 有一张很大的坐标纸。首先，他用一条线段将点 $(0, 4)$ 和点 $(2, 0)$ 连接起来，并将线段经过的 4 个单元格涂了颜色，如下图所示。接下来，Rodrigo 将点 $(2000, 3000)$ 与点 $(5000, 8000)$ 连接起来，并将线段与内部相交的单元格涂了颜色。这次他会涂多少个单元格？

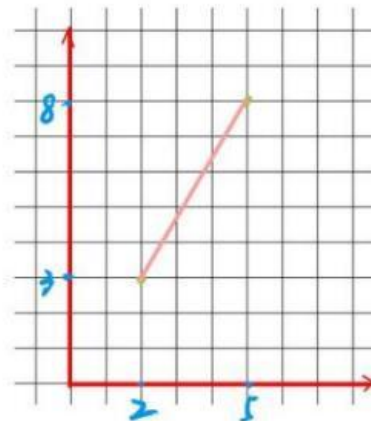


- (A) 6000 (B) 6500 (C) 7000 (D) 7500 (E) 8000

【答案】C

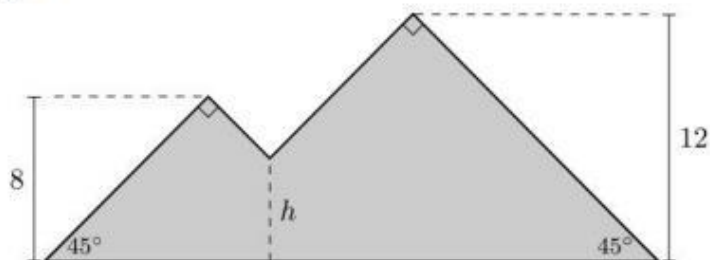
【解析】

可先作 $(2, 3)$ 、 $(5, 8)$
 观察得 该线段有 7 个方格
 $\Rightarrow 7 \times 1000 = 7000$



24. Jean made a piece of stained glass art in the shape of two mountains, as shown in the figure below. One mountain peak is 8 feet high and the other peak is 12 feet high. Each peak forms a 90° angle, and the straight sides of the mountains form 45° angles with the ground. The artwork has an area of 183 square feet. The sides of the mountains meet at an intersection point near the center of the artwork, h feet above the ground. What is the value of h ? (三角形的面积)

Jean 制作了一件山脉形状的彩色玻璃艺术品，如下图所示。一座山峰高 8 英尺，另一座山峰高 12 英尺。每个山峰形成了一个 90° 角，山脉的直角边与地面成 45° 角。艺术品的面积为 183 平方英尺。山脉的两侧在艺术品中部的一个交汇点上连接，该交汇点离地面 h 英尺高。 h 的值是多少？



- (A) 4 (B) 5 (C) $4\sqrt{2}$ (D) 6 (E) $5\sqrt{2}$

【答案】B

【解析】

如图，各边长皆可表示

$\Rightarrow S_1 + S_2 + S_3 + S_4 = 183$

梯形

$\Rightarrow \frac{1}{2} \times 8 \times 8 + \frac{1}{2} (h+8)(8-h) + \frac{1}{2} (h+12)(12-h) + \frac{1}{2} \times 12 \times 12 = 183$

$\Rightarrow (h+8)(8-h) + (12+h)(12-h) = 158$

$\Rightarrow 64 - h^2 + 144 - h^2 = 158$

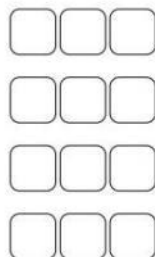
$h^2 = 25$

$h = 5$

25. A small airplane has 4 rows of seats with 3 seats in each row. Eight passengers have boarded the plane and are distributed randomly among the seats. A married couple is next to board. What is the probability there will be 2 adjacent seats in the same row for the couple?

一架小型飞机有 4 排座位，每排有 3 个座位。8 名乘客先登机后随机就坐。然后一对已婚夫妇随后登机。这对夫妇坐在同一排且相邻的概率是多少？

A small airplane has 4 rows of seats with 3 seats in each row. Eight passengers have boarded the plane and are distributed randomly among the seats. A married couple is next to board. What is the probability there will be 2 adjacent seats in the same row for the couple?



- (A) $\frac{8}{15}$ (B) $\frac{32}{55}$ (C) $\frac{20}{33}$ (D) $\frac{34}{55}$ (E) $\frac{8}{11}$

【答案】C

【解析】

1° 剩 4 位子为 一排 3 个 加 一排 1 个

此时 $P_1 = 1$

2° 两排分别 2 个

$$P_2 = \frac{C_4^2 \times A_2^2 \times 2 \times 2}{C_4^2 \times 3 \times 3} = \frac{8}{9}$$

3° 2, 1, 1 情形

3° 2, 1, 1 情形

$$N = 4 \times C_3^2 \times 3 \times 3 \times 3 = 324 \quad P_3 = \frac{2}{3}$$

4° 1, 1, 1, 1, $N = 3^4 = 81$, $P_4 = 0$

$$\Rightarrow P = \frac{36 \times 1 + 54 \times \frac{8}{9} + 324 \times \frac{2}{3}}{36 + 54 + 324 + 81} = \frac{20}{33}$$

	A	F
	C	E
	D	D
.	E	B